

SAFOD Model A: First-Order Tidal $\Delta v/v$ Benchmark

Where Every Number Comes From, What Is Assumed, and What Is Actually Being Calculated

Working theory notes for the 2026 SAFOD AWD experiment

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Contents

1	What Model A is, in one page	3
2	Parameter ledger: every numerical value used	3
3	What is <i>not</i> needed for Model A	4
4	Step 1: where the June 2026 tide shape comes from	4
4.1	External gravitational potential	4
4.2	Why degree 2 is enough for this benchmark	5
5	Step 2: why normalization makes Model A simpler	5
6	Step 3: where Niu's $2.4 \times 10^{-7} \text{ Pa}^{-1}$ comes from	6
6.1	What Niu actually measured	6
6.2	What “linear regression” means here	6
7	Step 4: where the 240 Pa comes from	7
8	Step 5: the two explicit Model A assumptions	7
8.1	Assumption 1: use Niu's stress scale to set our June 2026 amplitude	7
8.2	Assumption 2: transfer Niu's barometric coefficient to tidal stress	8
9	Step 6: substitute the actual numbers	8

10 What Model A does and does not justify	8
11 What a future successful tide experiment could do with this framework	9
12 Equation chain of custody for Model A	9
13 A compact group-meeting explanation	10
14 References	10

1 What Model A is, in one page

Model A is intentionally simple. It does *not* calculate a full 3-D tidal stress tensor at SAFOD depth. It combines three ingredients:

$$\boxed{\text{June 2026 degree-2 tide shape} \times \text{Niu tidal-stress scale} \times \text{Niu velocity-barometric-stress sensitivity}} \quad (1)$$

The model is

$$\tau(t) = \frac{W(t) - \overline{W}}{\max |W(t) - \overline{W}|}, \quad (2)$$

$$\Delta\sigma_A(t) = 240 \tau(t) \text{ Pa}, \quad (3)$$

$$\left(\frac{\Delta v}{v}\right)_A(t) = \left(2.4 \times 10^{-7} \text{ Pa}^{-1}\right) \Delta\sigma_A(t). \quad (4)$$

Therefore

$$\boxed{\left(\frac{\Delta v}{v}\right)_A(t) = 5.76 \times 10^{-5} \tau(t)} \quad (5)$$

and because $|\tau| \leq 1$,

$$\boxed{\max \left| \frac{\Delta v}{v} \right|_A = 5.76 \times 10^{-5} = 0.00576\%}. \quad (6)$$

The two important assumptions are: (1) use Niu et al.’s approximately 240 Pa SAFOD tidal-stress scale to set the amplitude of the June 2026 template; and (2) use Niu et al.’s barometrically derived velocity–stress coefficient as a first-order proxy for tidal stress sensitivity. Neither assumption is a direct measurement of the 2026 AWD mode’s tidal response.

2 Parameter ledger: every numerical value used

Quantity	Value	Where it enters / provenance
SAFOD latitude	35.9746°	June 2026 astronomical tide calculation; site coordinate used in notebook.
SAFOD longitude	−120.5516°	June 2026 astronomical tide calculation; site coordinate used in notebook.
Time window	2026-06-16 23:48 to 2026-06-17 23:44 UTC	AWD survey window.
Sampling	1 min	Numerical sampling of tide template.
G	6.674×10^{-11} SI	Standard gravitational constant used in tide potential.
R_E	6.371×10^6 m	Earth radius used in tide potential.

Moon mass	7.342×10^{22} kg	Standard astronomical input used in notebook.
Sun mass	1.989×10^{30} kg	Standard astronomical input used in notebook.
Degree retained	$n = 2$	Standard leading deformation-producing tide; Métivier & Conrad (2008) justify degree 2 as the dominant first-order luni-solar forcing.
Niu stress sensitivity	2.4×10^{-7} Pa $^{-1}$	Empirical linear-regression slope relating Niu’s SAFOD velocity variation to barometric stress.
Niu tidal-stress scale	~ 240 Pa	Niu et al.’s modeled solid-Earth tidal stress varied within about 240 Pa during their experiment.
Model A maximum $ \Delta v/v $	5.76×10^{-5} 0.00576%	= Product of the two Niu values when $ \tau = 1$.
Best 2026 AWD reliable level	0.5%	
		Empirical Deep-outbound injection-recovery sensitivity; used only for detectability comparison, not to construct Model A.

3 What is *not* needed for Model A

Model A does not require Young’s modulus, Poisson ratio, a plane-stress assumption, fault strike, fault dip, fault-normal traction, or a full tidal stress tensor. Those belong to the separate exploratory Model B calculation.

It also does not actually require Love-number amplitudes once the tide curve is normalized. Love numbers are useful for understanding the physical deformation, but the normalized Model A timing template can be constructed directly from the degree-2 tide-generating potential $W(t)$.

4 Step 1: where the June 2026 tide shape comes from

4.1 External gravitational potential

[**STANDARD THEORY**] For an external body b (Moon or Sun) with mass M_b and geocentric distance D_b , the Newtonian potential at a point $\mathbf{r}$ is

$$U_b(\mathbf{r}) = \frac{GM_b}{|\mathbf{D}_b - \mathbf{r}|}. \quad (7)$$

For $r/D_b < 1$, the potential can be expanded in Legendre polynomials:

$$U_b = \frac{GM_b}{D_b} \sum_{n=0}^{\infty} \left(\frac{r}{D_b} \right)^n P_n(\cos \theta_b). \quad (8)$$

The first three terms have different physical meanings. The $n = 0$ term is spatially constant. The $n = 1$ term corresponds to the nearly uniform acceleration of Earth as a whole toward the external body and is removed in an Earth-centered frame. The $n = 2$ term is the first term that produces differential gravity across Earth and therefore the leading deformation-producing tide.

At Earth’s surface, the degree-2 tide-generating potential is

$$W_b(t) = \frac{GM_b}{D_b(t)^3} R_E^2 P_2[\cos \theta_b(t)] \quad (9)$$

with

$$P_2(x) = \frac{1}{2}(3x^2 - 1). \quad (10)$$

The total first-order luni-solar forcing is

$$W(t) = W_{\text{Moon}}(t) + W_{\text{Sun}}(t). \quad (11)$$

4.2 Why degree 2 is enough for this benchmark

[**SOURCE – Métivier & Conrad (2008)**] Their body-tide treatment states that the degree-2 long-period, diurnal, and semidiurnal terms account for approximately 99.98% of the total luni-solar tide-generating-potential power. That is the justification for using degree 2 as a first-order timing template.

This does *not* mean degree 2 gives 99.98% of the exact local fault stress at SAFOD. It means it captures essentially all of the leading astronomical forcing power needed for the present detectability benchmark.

5 Step 2: why normalization makes Model A simpler

The earlier notebook converted $W(t)$ to a scalar surface areal-strain proxy,

$$\varepsilon_A(t) = C W(t), \quad C = \frac{2h_2 - 6l_2}{gR_E}. \quad (12)$$

For the PREM Love numbers used previously, $C > 0$. Model A then mean-removes and normalizes the curve:

$$\tau(t) = \frac{\varepsilon_A(t) - \overline{\varepsilon_A}}{\max |\varepsilon_A(t) - \overline{\varepsilon_A}|}. \quad (13)$$

Substitute $\varepsilon_A = CW$:

$$\tau(t) = \frac{CW(t) - C\overline{W}}{\max |CW(t) - C\overline{W}|} \quad (14)$$

$$= \frac{C[W(t) - \overline{W}]}{|C| \max |W(t) - \overline{W}|}. \quad (15)$$

Because $C > 0$,

$$\tau(t) = \frac{W(t) - \overline{W}}{\max |W(t) - \overline{W}|}. \quad (16)$$

Important simplification: the Love-number amplitude, g , and R_E cancel out of the normalized timing template. Model A needs the degree-2 astronomical time dependence, not the absolute surface-strain amplitude.

The normalization forces

$$-1 \leq \tau(t) \leq 1, \quad (17)$$

so τ contains timing and relative shape only.

6 Step 3: where Niu’s $2.4 \times 10^{-7} \text{ Pa}^{-1}$ comes from

6.1 What Niu actually measured

[SOURCE – Niu et al. (2008)] Niu et al. repeatedly measured travel-time/velocity changes with an active source at SAFOD. After correcting the delay-time series for temperature, they found that the velocity variation tracked barometric-pressure changes.

They report a representative velocity perturbation of order

$$\left| \frac{\Delta v}{v} \right| \sim 3 \times 10^{-4} \quad (18)$$

associated with a barometric-pressure change of about

$$|\Delta P_{\text{atm}}| \sim 1.3 \text{ kPa} = 1.3 \times 10^3 \text{ Pa}. \quad (19)$$

The ratio of those two rounded numbers is

$$\frac{3 \times 10^{-4}}{1.3 \times 10^3 \text{ Pa}} \approx 2.31 \times 10^{-7} \text{ Pa}^{-1}, \quad (20)$$

which is close to their reported coefficient but is not the formal estimate.

6.2 What “linear regression” means here

Niu had many paired observations of barometric pressure and measured velocity variation. Conceptually, write

$$y_i = \left(\frac{\Delta v}{v} \right)_i, \quad x_i = \Delta P_i. \quad (21)$$

Fit the straight-line model

$$\hat{y}_i = b + S_{\text{baro}} x_i. \quad (22)$$

Least squares chooses b and S_{baro} to minimize

$$\sum_i [y_i - (b + S_{\text{baro}} x_i)]^2. \quad (23)$$

The fitted slope is

$$S_{\text{baro}} \approx 2.4 \times 10^{-7} \text{ Pa}^{-1}. \quad (24)$$

Its meaning is concrete: for the seismic observable Niu measured, the best-fit linear relationship corresponds to roughly 2.4×10^{-7} fractional velocity change per pascal of barometric loading.

For example, for a 1 kPa change,

$$(2.4 \times 10^{-7} \text{ Pa}^{-1})(1000 \text{ Pa}) = 2.4 \times 10^{-4} = 0.024\%. \quad (25)$$

Provenance: $2.4 \times 10^{-7} \text{ Pa}^{-1}$ is an empirical coefficient from Niu’s barometric-loading regression. It is not a theoretical elastic constant and it is not a directly measured tidal sensitivity.

7 Step 4: where the 240 Pa comes from

[**SOURCE – Niu et al. (2008)**] Niu et al. also calculated the solid-Earth tidal stress during their observing interval and state that it “varies within 240 Pa.” For Model A we retain

$$\Delta\sigma_{\text{tide}} \sim 240 \text{ Pa} \quad (26)$$

as a characteristic SAFOD tidal-stress scale.

The wording does not, by itself, unambiguously specify whether 240 Pa should be treated as a maximum absolute amplitude, half range, or peak-to-peak excursion of a particular stress component. Therefore Model A should not pretend that 240 Pa is an exact June 2026 amplitude.

8 Step 5: the two explicit Model A assumptions

8.1 Assumption 1: use Niu’s stress scale to set our June 2026 amplitude

Our normalized tide $\tau(t)$ has no units and has maximum absolute value 1. We construct

$$\Delta\sigma_A(t) = 240 \tau(t) \text{ Pa}. \quad (27)$$

This is **our modeling choice**. Niu did not publish this June 2026 stress time series. The purpose is to combine our June astronomical timing with a site-relevant stress scale from their SAFOD experiment.

8.2 Assumption 2: transfer Niu’s barometric coefficient to tidal stress

We then assume, to first order,

$$S_{\text{tide}} \approx S_{\text{baro}}. \quad (28)$$

This gives

$$\left(\frac{\Delta v}{v}\right)_A(t) = S_{\text{baro}} \Delta \sigma_A(t). \quad (29)$$

This is also **our modeling assumption**. Barometric loading and tidal loading need not have identical effects on the relevant rock volume or seismic mode. The coefficient is therefore a transfer assumption used to obtain a first-order scale.

9 Step 6: substitute the actual numbers

Start with

$$\Delta \sigma_A(t) = 240\tau(t) \text{ Pa} \quad (30)$$

and

$$S_{\text{baro}} = 2.4 \times 10^{-7} \text{ Pa}^{-1}. \quad (31)$$

Then

$$\left(\frac{\Delta v}{v}\right)_A(t) = (2.4 \times 10^{-7} \text{ Pa}^{-1}) (240\tau(t) \text{ Pa}) \quad (32)$$

$$= 5.76 \times 10^{-5} \tau(t). \quad (33)$$

Thus

$$\boxed{\max |\Delta v/v|_A = 5.76 \times 10^{-5} = 0.00576\%}. \quad (34)$$

The 2026 Deep-outbound injection-recovery result reliably recovered a tested 0.5% apparent-velocity change. The ratio is

$$\frac{0.5\%}{0.00576\%} \approx 87. \quad (35)$$

So the Model A characteristic tidal scale is about 87 times smaller than the best demonstrated reliable AWD change level.

10 What Model A does and does not justify

Model A **does** justify an order-of-magnitude statement: a Niu-calibrated SAFOD tidal velocity response is expected to be of order 10^{-5} – 10^{-4} in fractional $\Delta v/v$, well below the current AWD reliable-detection level.

Model A **does not** independently predict the exact June 2026 fault-normal tidal stress, the exact sign or phase of the true AWD-mode response, or the true constitutive sensitivity of the Nano or Deep guided modes.

The safest description is:

Model A is a SAFOD-calibrated first-order detectability benchmark. The June 2026 degree-2 tide provides the timing and relative shape, while Niu et al.’s modeled SAFOD tidal-stress scale and empirically measured velocity–barometric-stress sensitivity provide the response scale.

11 What a future successful tide experiment could do with this framework

A future experiment with much better repeatability and a much longer record could measure

$$\left(\frac{\Delta v}{v}\right)_{\text{obs}}(t) \quad (36)$$

and compare it with an independently modeled tidal predictor.

Instead of borrowing Niu’s barometric coefficient, the future experiment could estimate its own tidal coefficient, for example

$$\left(\frac{\Delta v}{v}\right)_{\text{obs}}(t) = b + S_{\text{tide}}\Delta\sigma_{\text{tide}}(t) + \eta(t). \quad (37)$$

Then S_{tide} would be measured directly for that seismic observable. Comparing it with Niu’s barometric coefficient would test whether the system responds similarly to different kinds of loading.

12 Equation chain of custody for Model A

Equation/value	Provenance	Interpretation
$U_b = GM_b/ \mathbf{D}_b - \mathbf{r} $	Standard Newtonian gravity	Starting potential of Sun/Moon.
Legendre expansion	Standard mathematics / tide theory	Separates spatial degrees.
Keep $n = 2$	Standard tide theory	Leading deformation-producing term.
99.98% degree-2 power	Métivier & Conrad (2008)	Justifies first-order degree-2 forcing.
$W_b = GM_b D_b^{-3} R_E^2 P_2$	Standard degree-2 tide potential	Generates June 2026 timing.
$\tau = (W - \bar{W})/\max W - \bar{W} $	Modeling choice	Removes amplitude and retains timing/shape.
$S_{\text{baro}} = 2.4 \times 10^{-7} \text{ Pa}^{-1}$	Niu et al. (2008)	Empirical barometric velocity sensitivity.
Tidal stress within $\sim 240 \text{ Pa}$	Niu et al. (2008)	Site-relevant stress scale for their interval.

$\Delta\sigma_A = 240\tau$	Modeling choice	Imposes Niu’s stress scale on June 2026 timing.
$(\Delta v/v)_A = S_{\text{baro}}\Delta\sigma_A$	Transfer assumption	Treats barometric coefficient as first-order tidal proxy.
5.76×10^{-5}	Arithmetic from the two Niu values	Maximum scale under the normalization convention.

13 A compact group-meeting explanation

For the first-order benchmark I only need the degree-2 Sun–Moon tide to get the June 2026 timing. I normalize that curve, so the Love-number amplitude cancels out. I then use two SAFOD-specific quantities from Niu et al.: their modeled tidal-stress scale of about 240 Pa and their empirically fitted barometric velocity–stress sensitivity of $2.4 \times 10^{-7} \text{ Pa}^{-1}$. Multiplying those gives a characteristic $\Delta v/v$ scale of 5.76×10^{-5} , or about 0.006%. Applying the barometric coefficient to tidal loading is an explicit transfer assumption, so I treat this as a detectability benchmark rather than an exact prediction.

14 References

References

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